

Project Design Phase-II Technology Stack (Architecture & Stack)

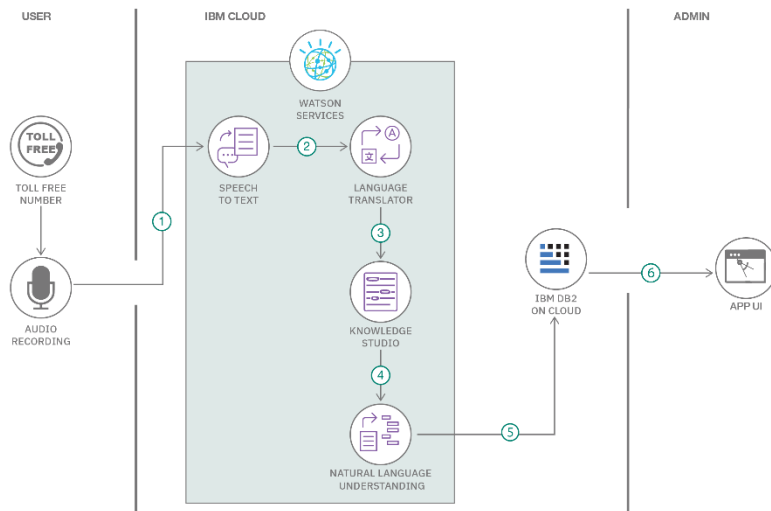
Date	07 July 2026
Team ID	
Project Name	Qcab – Online Cab Booking Application
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application through the web browser (Home, Book Cab, My Booking, Admin panel)	HTML, CSS, JavaScript, React.js, Bootstrap
2.	Application Logic-1	Core booking logic — registration, login, cab search & booking, fare calculation and driver assignment	Node.js, Express.js
3.	Application Logic-2	Authentication and authorization for users and admins	JWT, bcrypt (Node.js / Express)
4.	Application Logic-3	Admin operations — manage users, cabs and bookings	Node.js, Express.js (REST APIs)
5.	Database	Stores users, cabs and booking records	MongoDB (NoSQL)
6.	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7.	File Storage	Storage for uploaded car images	Local filesystem via Multer
8.	External API-1	Not used — pickup and drop locations are entered manually by the user	Not applicable
9.	External API-2	Not used in the current build	Not applicable
10.	Machine Learning Model	Not used in the current build	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on local system Local Server Configuration: Node.js / Express server on localhost with the React dev server; MongoDB running locally Cloud Server Configuration: Not applicable (local deployment)	Local server (Node.js + MongoDB)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks and libraries used to build Qcab	React.js, Node.js, Express.js, MongoDB, Bootstrap
2.	Security Implementations	Access control and data protection implemented in the application	Password hashing with bcrypt, JWT-based authentication, role-based access (user / admin)
3.	Scalable Architecture	3-tier MERN architecture (React client, Express/Node API layer, MongoDB data layer) so each layer can scale independently	MERN 3-tier, component-based React
4.	Availability	Runs on a local server for the demo; the stateless REST API and MongoDB Atlas allow cloud deployment for higher availability	Node.js server (local); cloud-deployable
5.	Performance	React client-side rendering, lightweight JSON REST responses and indexed MongoDB queries keep the app responsive	React, Express, MongoDB indexing

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>